

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	RISHIK JK	Reg. No.:	192511052
Date:	25-08-2026	Duration:	60 minutes

Code of Conduct

I RISHIK JK (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Rishik JK

CO4-AT1-Test Case Design Assignment

NAME: JK RISHIK

SLOT: D

REGNO :192511052

COURSE CODE: CSA1007

DATE:25/08/2026

Problem statement :Smart Renewable Battery Recycling Platform

1. Introduction

The Smart Renewable Battery Recycling Platform is developed to manage battery inventory, recycling activities, hazardous conditions, logistics, sustainability reports, and operator notifications. The purpose of this test case document is to verify whether the system works correctly under normal, invalid, boundary, and exceptional conditions.

2. Functional Requirements to be Tested

The main functional requirements are:

- User login and authentication.
- Battery inventory management.
- Recycling task management.
- Hazard detection and alert generation.
- Logistics tracking.
- Sustainability report generation.
- Notification management.

3. Non-Functional Requirements to be Tested

The main non-functional requirements are:

- **Performance:** The system should respond quickly to user requests.
- **Security:** Only authorized users should access the system.
- **Reliability:** The system should work without data loss or frequent failure.
- **Usability:** The interface should be simple and easy to use.

- **Availability:** The system should be available whenever required.
- **Scalability:** The system should support increasing users and battery records.

4. Test Scenarios

The following major test scenarios are considered:

- Verify valid and invalid user login.
- Verify addition and update of battery details.
- Verify inventory quantity limits.
- Verify recycling task assignment and status update.
- Verify detection of hazardous battery conditions.
- Verify alert generation.
- Verify logistics tracking and shipment status.
- Verify sustainability report generation.
- Verify system response for invalid or missing data.

5. Detailed Test Cases:

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result
TC01	Valid User Login	Enter valid username and password and click Login	Username: operator1, Password: Op@123	User should login successfully and dashboard should open	As Expected
TC02	Invalid User Login	Enter incorrect password and click Login	Username: operator1, Password: wrong123	System should display invalid credentials message	As Expected
TC03	Empty Login Fields	Leave username and password empty and click	Blank fields	System should ask the user to enter required details	As Expected

		Login			
TC0 4	Add Battery Record	Enter valid battery details and click Save	Type: Li-ion, Qty: 20, Condition: Used	Battery record should be stored successfully	As Expect
TC0 5	Invalid Battery Quantity	Enter a negative battery quantity	Qty: -5	System should reject the value and display an error	As Expect
TC0 6	Boundary Battery Quantity	Enter minimum valid quantity	Qty: 1	System should accept the battery quantity	As Expect
TC0 7	Update Inventory	Select a battery record and update quantity	Existing: 20, Updated: 25	Inventory should display updated quantity	As Expect
TC0 8	Assign Recycling Task	Manager assigns a valid recycling task	Task: Sort Li- ion batteries	Task should be assigned to the selected operator	As Expect
TC0 9	Update Task Status	Operator changes task status	Pending → Completed	System should update the task status successfully	As Expect

6. Test Design Techniques

6.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

For battery quantity:

- Valid class: Quantity greater than or equal to 1.
- Invalid class: Quantity less than 1.

Example:

Quantity = 20 → Valid

Quantity = -5 → Invalid

6.2 Boundary Value Analysis

Boundary Value Analysis checks values near the minimum or maximum allowed limit.

For battery quantity:

- 0 → Invalid
- 1 → Valid
- 2 → Valid

This helps verify whether the system correctly handles values close to the boundary.

6.3 Decision Table Testing

Decision Table Testing is used for login verification.

6.4 State Transition Testing

State Transition Testing is used for recycling task status.

Task States:

Pending → In Progress → Completed

A task should move only through valid states. For example, a completed task should not return directly to the pending state without proper authorization.

7. Requirement Traceability and Coverage

The test cases are mapped with the main system requirements to ensure proper coverage.

- **Login and Security:** TC01, TC02, TC03, TC18
- **Battery Inventory:** TC04, TC05, TC06, TC07
- **Recycling Operations:** TC08, TC09
- **Hazard Detection:** TC10, TC11, TC12, TC17
- **Logistics Tracking:** TC13, TC14
- **Sustainability Reports:** TC15, TC16

The test cases cover normal conditions, invalid inputs, boundary values, missing data, unauthorized access, and exceptional situations.

8. Types of Testing Applied

Different types of testing are used to check whether the Smart Renewable Battery Recycling Platform works correctly and satisfies user requirements.

8.1 Functional Testing

Functional testing is used to verify the main functions of the system. It checks features such as user login, battery inventory, recycling task management, hazard detection, logistics tracking, report generation, and notifications.

8.2 Validation Testing

Validation testing checks whether the user enters correct and acceptable data into the system. For example, battery quantity should not contain negative values, and mandatory fields should not be left empty.

8.3 Security Testing

Security testing is performed to ensure that unauthorized users cannot access restricted features. For example, a recycling operator should not be able to access administrator functions.

8.4 Performance Testing

Performance testing checks whether the system responds quickly while processing inventory records, recycling tasks, reports, and other user requests.

8.5 Usability Testing

Usability testing ensures that the application is simple and easy to understand. Users should be able to access important features without confusion.

9. Testing Process

The testing process is carried out step by step to identify errors and verify the overall working of the system.

Step 1 – Requirement Analysis

The functional and non-functional requirements of the system are studied to understand what features need to be tested.

Step 2 – Test Scenario Identification

Different situations such as valid login, invalid login, battery entry, hazard detection, logistics tracking, and report generation are identified.

Step 3 – Test Case Preparation

Test cases are prepared with test data, test steps, expected results, and actual results.

Step 4 – Test Execution

The prepared test cases are executed on the system and the actual output is compared with the expected output.

Step 5 – Defect Identification

If the actual result is different from the expected result, the issue is identified and recorded as a defect.

Step 6 – Retesting

After correcting the identified defect, the same test case is executed again to confirm that the problem has been solved.

10. Defect Handling and Test Result Analysis

During software testing, defects may occur when the actual output of the system does not match the expected output. These defects should be identified and corrected before the system is released.

Defects can be classified based on their impact:

- **Critical:** The system or an important feature cannot function.
- **High:** A major functionality does not work correctly.
- **Medium:** The functionality works but contains some errors.
- **Low:** Minor issues that do not affect the main operation of the system.

After fixing a defect, retesting should be performed to ensure that the problem has been resolved. Regression testing can also be performed to verify that the changes have not affected other existing features.

The final test results help determine whether the Smart Renewable Battery Recycling Platform is stable and ready for use.

8. Conclusion

The prepared test cases verify the important functions of the Smart Renewable Battery Recycling Platform. The test design covers login, inventory management, recycling operations, hazard detection, logistics tracking, reporting, and notifications. Techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing are applied to improve test coverage and system reliability.